

Project Proposal: AI Wardrobe Stylist

Course: CSCI 4397 - Software Engineering with AI

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Problem

People every day wear a small part of everything that they own in their wardrobe because remembering what's clean, what matches, and what hasn't been worn lately (or at all) is hard to track by memory. This leads to spending time every morning and underused clothes.

Proposed Solution

A web app that digitizes a user's closet through photo uploads, using AI vision tagging to automatically classify each item, then generates a daily outfit suggestion based on weather, laundry status, and personal style. A short onboarding quiz seeds the recommendation engine with initial preferences (solving the cold-start problem), and the system improves over time through user feedback on suggestions.

MVP Scope

Included in MVP

Feature

- Photo upload with AI auto-tagging (type, color, season)
- Closet inventory view
- Mark item clean / in laundry
- Multi-user profiles under one household
- Daily outfit suggestion (weather + clean + repeat avoidance)
- Cold-start quiz on first use
- Thumbs up / down feedback on suggestions

Why it's core

- Without this, the whole "no manual entry" premise falls apart
- Basic usability, need to see what's there
- Directly affects what can be suggested
- Central to how you'll actually use it
- The core value proposition
- Seeds personalization before any swipe data exists
- How the system actually learns over time

- **Stretch (attempt after MVP is solid)**

- Color coordination / pairing rules for genuinely stylish combinations
- Layering logic for cold weather (base layer + outer layer suggestions)
- Outfit history log ("what did I wear last Tuesday")
- Style vibe drift over time (taste changes season to season)

- **Tech Stack (all free tier)**

- **Frontend:** React, hosted free on Vercel or Netlify
- **Backend / DB:** Supabase (Postgres + auth + file storage, free tier)
- **Vision tagging:** A free-tier multimodal model (e.g. Gemini) for garment classification from photos
- **Weather:** OpenWeatherMap free tier
- **Recommendation logic:** Rule-based scoring to start, blended with the preference vector from the quiz + feedback loop (see below)

- **High-Level Data Model**

- users (household_id, name, quiz_preferences)
- garments (user_id, image_url, type, color, season, warmth, status: clean/dirty, last_worn_date)
- preference_vector (user_id, tag, weight) — seeded by quiz, updated by feedback
- suggestions (user_id, date, garment_ids, feedback: up/down/null)

Expected Outcome

A working, demoable web app that solves a genuine daily problem, with real AI engineering depth in vision tagging, cold-start personalization, and preference learning.